



THE STATE UNIVERSITY OF ZANZIBAR (SUZA)
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY

Project Name: Citizen Alert System

Version:1.0

Date: February 2026

Author(s):

Zahor Khalfan Omar (24BCS018)

Khadija Ali Said (24BCS026)

Abdulkadir Bakar Mbwana (24BCS028)

ACADEMIC YEAR 2025/2026

Outline

- 1 Introduction
- 2 Solution
- 3 Demo
- 4 Development Process
- 5 Testing
- 6 Challenges & Lessons
- 7 Future Work
- 8 Conclusion

Project Overview

Project Name

Citizen Alert System

What problem are we solving?

Citizens in Zanzibar do not have a reliable way to receive official government alerts. Important information spreads through WhatsApp and social media where fake news is common.

Why is this problem important?

During emergencies like floods or health crises, citizens need fast and verified information. Without a proper system, people miss important announcements or receive wrong information.

Who faces this problem?

- Citizens of Zanzibar who need alerts about health, education, and safety
- Government ministries trying to reach specific groups of people
- ICT department that needs to manage who can send official alerts

Objectives

Primary Objective:

Build a command-line system where government offices can send targeted alerts to citizens based on their region, job status, and education level.

Secondary Objectives

1. Allow citizens to register with their personal information
2. Enable different government offices to create and manage their own alerts
3. Give Super Admin (ICT Department) control over all office accounts
4. Sort alerts by priority so emergency messages appear first

Target Beneficiaries: Citizens of Zanzibar, Government Ministries and Public Offices, Department of ICT - Zanzibar

Target Users

Primary Users:

- **Citizens of Zanzibar** - Register to receive alerts about health, education, emergencies, and public services
- **Government Admins** - Create and send announcements to specific citizen groups
- **Super Admin (ICT Department)** - Manage all office accounts and oversee the entire system

User Needs:

- Citizens need alerts that match their region and interests only
- Government offices need to target specific groups like students, workers, or people in certain regions
- Emergency alerts must reach everyone regardless of their preferences or location

Proposed Solution

How We Solved It

We built a **command-line system in C language** that allows government offices to create alerts and citizens to receive only the alerts that match their profile. The system uses linked lists to store data in memory and a queue to manage notifications. Alerts are automatically sorted by priority so emergency messages always appear first.

Key Features

Feature 1: Role-Based Access

- Super Admin (ICT Department) can create, delete, and manage all office accounts
- Regular Admins (Government Ministries) can only manage their own office's alerts

Feature 2: Targeted Alerts

Government offices can send alerts to four different audience types:

- All citizens
- Citizens in a specific region (like Unguja or Pemba)
- Citizens with a specific status (Student, Worker, or General)
- Citizens with a minimum education level (Primary to Postgraduate)

Feature 3: Priority System

Alerts have four priority levels:

- Level 0: Low priority
- Level 1: Normal priority
- Level 2: High priority
- Level 3: EMERGENCY (bypasses all targeting rules)

Feature 4: Citizen Registration

Citizens register with their name, region, status, and education level. The system gives them a unique ID number starting from 100. They use this ID to view their alerts.

Feature 5: Alert Filtering

When a citizen views alerts, the system automatically shows only the alerts that match their:

- Region (if alert targets a specific region)
- Status (if alert targets students, workers, or general public)
- Education level (if alert requires minimum education)

Feature 6: Emergency Override

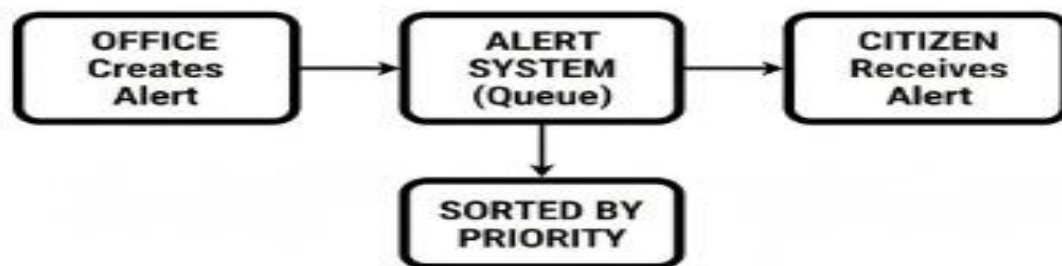
Emergency priority alerts are shown to EVERY citizen, no matter their region, status, or education level.

System Architecture

Frontend: Command-line interface (terminal)

Backend: C programming language with linked lists and queue data structures

Database: No external database - data stored in memory using structs and linked lists



Technology Stack

Programming Language:

C Language - Core system development

Data Structures Used:

Linked Lists - Store office accounts, citizens, and notifications

Queue - Manage alerts in FIFO order with priority sorting

Development Tools:

GCC Compiler - Compile C code into working program

VS Code - Write and edit code

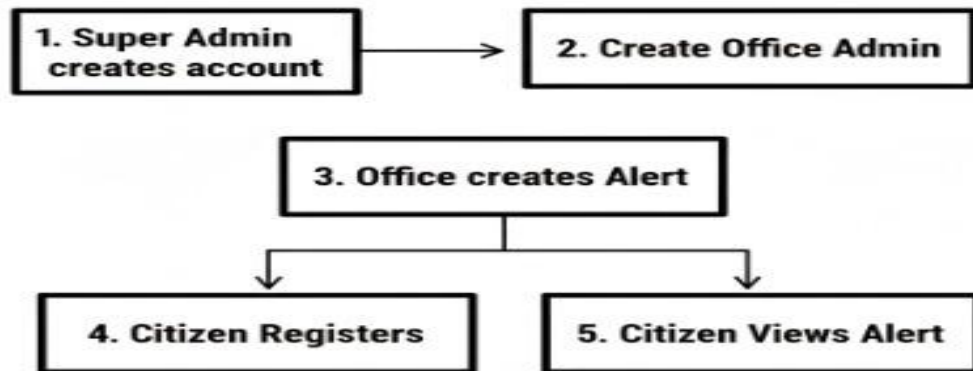
Git/GitHub - Save and share code with team

Terminal - Run and test the program

Live Demo

What We Will Show

1. Office Login: Super Admin creates account
2. Create Office Account: Add new government office
3. Create Alert: Office sends message with priority
4. Citizen Register: New citizen signs up
5. View Alerts: Citizen sees their alerts



System in Action

Main Menu

```
CITIZEN ALERT SYSTEM v5.0
Zanzibar

Login as: [0] Office Admin  [1] Citizen  [-1] Quit
Choice: 0
```

Super Admin Menu

```
Welcome, Admin! [eGAZ | Super Admin]

Office Menu [SUPER ADMIN]
-- Notifications --
1. Create Notification
2. List Notifications
3. Search Notification
4. Remove Notification
-- Account Management --
5. Create Office Account
6. List Office Accounts
7. Toggle Account Active
8. Delete Office Account
9. Reset Account Password
0. Logout
```

Creating an Alert

```
Welcome, IT! [ZECO | Admin]

Office Menu [ZECO]
1. Create Notification
2. List Notifications
3. Search Notification
4. Remove Notification
0. Logout

Choice: 1
Title      : Maboresho ya Uneme
Message    : Uneme utazimika siku ya Jumamosi ya tarehe 28/2/2026 kuanzia saa 2:00 asubuhi mpaka saa 10:00 jioni samahanini kwa usumbufu utakaojitokeza.
Priority [0=Low 1=Normal 2=High 3=Emergency]: 3
Target [0=All 1=Region 2=Status 3=Education]: 0
[OK] Notification created. ID = 1
```

Citizen viewing Alerts

```
Citizen Portal
1. Register
2. Search Citizen
3. View My Alerts
0. Exit

Choice: 3
Enter your Citizen ID: 100

Alerts for Khalsa:

[EMERGENCY] Maboresho ya Uneme
From: ZECO | 2026-02-25 12:24
Uneme utazimika siku ya Jumamosi ya tarehe 28/2/2026 kuanzia saa 2:00 asubuhi mpaka saa 10:00 jioni samahanini kwa usumbufu utakaojitokeza.
```

List of Notification

```
Login as: [0] Office Admin  [1] Citizen  [-1] Quit
Choice: 0
Username : IT
Password : IT012

Welcome, IT! [ZECO | Admin]

Office Menu [ZECO]
1. Create Notification
2. List Notifications
3. Search Notification
4. Remove Notification
0. Logout

Choice: 2
```

ID	Title	Office	Priority	Target
1	Maboresho ya Umeme	ZECO	EMERGENCY	All

GitHub Repository: https://github.com/zvhor/citizen_alert_system.git

Demo Available: On request

Methodology (How We Built It)

Our Development Approach

We used **Agile/Scrum** methodology to build this project.

What does that mean?

Instead of building everything at once, we worked in small cycles called "sprints". Each sprint was 2 weeks long. At the end of each sprint, we had something working to show.

Why We Chose Agile/Scrum

Easy to track progress - We knew exactly what to finish each sprint

Flexible - We could change priorities if something wasn't working

Team friendly - Each person had clear tasks

Less risk - If something failed, we only lost 2 weeks of work

Sprint Details

Sprint Duration: 2 weeks per sprint

Total Sprints: 4 sprints

Total Project Duration: 8 weeks

Sprint Breakdown

Sprint 1: Setup and Structure

- Created data structures
- Set up GitHub
- Basic menu system

Sprint 2: Office Functions

- Office login system
- Create notifications
- List notifications

Sprint 3: Citizen Functions

- Citizen registration
- View alerts by priority
- Search citizens

Sprint 4: Admin Features

- Super Admin controls
- Account management
- Testing and fixes

Team Role

Name	Role	Student ID	Responsibilities
Zahor Khalfan Omar	Project Manager	24BCS018	• Led the team • Planned sprints • Managed timeline • Coordinated between members
Khadija Ali Said	System Analyst	24BCS026	• Designed system structure • Planned features • Analyzed user needs • Tested the system
AbdulKadir Bakar Mbwana	Developer	24BCS028	• Designed system structure • Planned features • Analyzed user needs • Tested the system

Challenges & Lessons Learned

Challenges

Challenge 1: Data Storage

Problem: How to store all information without a real database

Solution: Used Linked Lists in C to store offices, citizens, and alerts in memory

Challenge 2: Sorting Alerts by Priority

Problem: Emergency alerts must show first, but new alerts go to the end

Solution: Created `sort_by_priority()` function that rearranges alerts after each new one

Challenge 3: Targeting Right Citizens

Problem: Each alert should only go to specific people (region, status, education)

Solution: Added `matches()` function that checks each citizen against alert rules

Lessons Learned

What We Learned From This Project

Technical Lessons

- How to use linked lists to store data in C
- How to sort data using bubble sort algorithm
- How to filter information based on user criteria
- How to manage memory properly to avoid crashes

Teamwork Lessons

- Communication is key - we used WhatsApp daily
- Dividing work by roles made things faster
- Helping each other when someone faced problems
- Meeting deadlines by planning sprints properly

Project Management Lessons

- Start with simple features, then add complex ones
- Test early, test often - find bugs quickly
- Document code so others understand it
- Backup work on GitHub in case computer fails

Future Improvements

What We Can Add Next

Short-term Improvements

- Save data to files - So information is not lost when program closes
- Add more citizen details - Phone numbers for SMS alerts
- Better search - Search alerts by date or keyword

Long-term Plans

- Web Version - Move from C program to website everyone can access
- Mobile App - Citizens get alerts on their phones
- SMS Integration - Send alerts via text message
- Multiple Languages - Swahili and English options

Future Vision

- Today: C Program (Terminal Only)
- Next: Save to Files
- Future: Website + Mobile App + SMS
- Goal: Every citizen gets alerts anywhere, anytime

Summary

What We Built

A C program that lets government offices send targeted alerts to citizens based on their location, status, and education level.

Key Achievements

- Created working alert system in C language
- Built priority system (Emergency alerts show first)
- Added targeting (send to specific regions or groups)
- Supported multiple government offices
- Citizens can register and get relevant alerts

Impact

- Government: Can reach citizens quickly during emergencies
- Citizens: Receive only alerts that matter to them
- Zanzibar: Better communication = safer communities

Our information

GitHub: https://github.com/zvhor/citizen_alert_system.git

Email: khadija09said09@gmail.com

Email:abdulkadirbakar39@gmail.com

Email:labszahor@gmail.com

Our Team

Zahor Khalfan Omar - Project Manager (24BCS018)

Khadija Ali Said - System Analyst (24BCS026)

AbdulKadir Bakar Mbwana - Developer (24BCS028)